A1 Q4a Edward Tamurkov 20401544

Tuesday, October 6, 2026 3:12 PM

a) $\nabla\nabla_{12}X_t = \nabla(X_t - X_{t-12})$
$= X_t - X_{t-1} - (X_{t-12} - X_{t-13})$
$= (\alpha + \cancel{\beta t} + \cancel{S_t} + Y_t)$
$- (\alpha + \cancel{\beta(t-1)} + \cancel{S_{t-1}} + Y_{t-1})$
$- (\alpha + \cancel{\beta(t-12)} + \cancel{S_{t-12}} + Y_{t-12})$
$+ (\alpha + \cancel{\beta(t-13)} + \cancel{S_{t-13}} + Y_{t-13})$
$= Y_t - Y_{t-1} - Y_{t-12} + Y_{t-13}$

$\nabla\nabla_{12}X_t$ is stationary b/c $E[\nabla\nabla_{12}X_t] = \overbrace{1-1-1+1}^{\text{coefficients of } Y_t} = 0$
and $Cov(\nabla\nabla_{12}X_{t+h}, \nabla\nabla_{12}X_t)$ = some combination of $\gamma_Y(h)$ which clearly doesn't depend on t.

ACVF
By property 4 of covariance:

$\gamma(h) = \sum_a \sum_b c_a c_b Cov(Y_{t+h-a}, Y_{t-b})$
$= \sum_a \sum_b c_a c_b \gamma_Y(h-a+b)$ by linearity of Cov., $\overbrace{(t+h-a)-(t-b)}^{\text{lag subscripts of } Y} = h-a+b$

For $a, b \in \{0, 1, 12, 13\}$ (the Y_t lags above),
and c_i is the coefficient (-1 or 1) for the corresponding Y_i for $i = a$ or b

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b) $\nabla_{12}^2 X_t = \nabla_{12}(X_t - X_{t-12})$
$= X_t - X_{t-12} - (X_{t-12} - X_{t-24})$
$= (a - bt)\varepsilon_t + Y_t$
$- (a + b(t-12))\varepsilon_{t-12} + Y_{t-12}$
$- (a + b(t-12))\varepsilon_{t-12} + Y_{t-12}$
$+ (a + b(t-24))\varepsilon_{t-24} + Y_{t-24}$
$= Y_t - 2Y_{t-12} + Y_{t-24}$

$\nabla_{12}^2 X_t$ is stationary b/c $E[\nabla_{12}^2 X_t] = \overbrace{1 - 2 + 1}^{\text{coefs. of } Y_t} = 0$
and $\text{Cov}(\nabla_{12}^2 X_{t+h}, \nabla_{12}^2 X_t)$ = some combination of $\gamma_Y(h)$ which clearly doesn't depend on t.

ACVF

By property 4 of covariance:

$\gamma(h) = \sum_a \sum_b c_a c_b \text{Cov}(Y_{t+h-a}, Y_{t-b})$
$= \sum_a \sum_b c_a c_b \gamma_Y(h - a + b)$ by linearity of Cov., $\overbrace{(t+h-a) - (t-b) = h - a + b}^{\text{lag subscripts of } Y}$

For $a, b \in \{0, 12, 24\}$ (the Y_t lags above), and c_i is the coefficient $(-2, -1, 1)$ for the corresponding Y_i for $i = a$ or b.